

TrES-2b

R-0085

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_181023 · created 2026-07-15T21:08:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458414.62142 +/- 0.00063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.044
Transit depth (Rp/Rs)^2	2.06 +/- 1.25 [%]
Orbital Inclination (inc)	87.27 +/- 2.9
Airmass coefficient 1 (a1)	2.745 +/- 0.097
Airmass coefficient 2 (a2)	-0.75 +/- 0.36
Scatter in the residuals of the lightcurve fit is	3.25 %
Best Comparison Star	None
Optimal Aperture	3.99
Optimal Annulus	5.78
Transit Duration (day)	0.084 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.144 $\pm$ 0.044 vs 0.1254 (deviation 0.42 σ)
Duration fit vs published	0.084 d vs 0.0746 d (deviation 0.52 σ)
Mid-time O–C	-0.9 min
Depth SNR	3.3
Residual scatter	3.25 %

Light curve

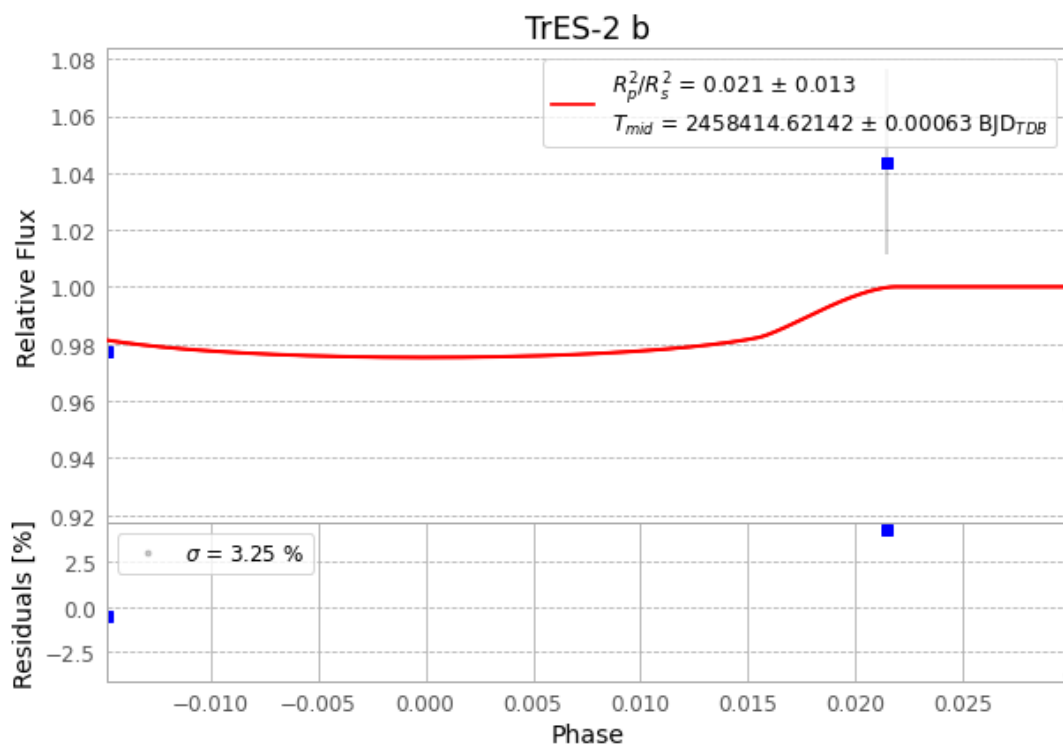


Figure 1 — FinalLightCurve_TrES-2 b_2018-10-22.png (SHA-256 b8fc5f0b0e7561a4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 341], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c79f3103e084aea7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

54 FITS frames (35 MB), TRES-2181023020012.FITS → TRES-2181023044212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-181023.log (976f1bc93d79...), FinalLightCurve_TrES-2 b_2018-10-22.png (b8fc5f0b0e75...), FinalLightCurve_TrES-2 b_2018-10-22.csv (880f944d0f3d...)

Compute resources

Wall time 195.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 21:08Z.